

ADA: Demographic Methods

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Week 7: Kinship

Overview

- ▶ What is kinship demography
- ▶ Goodman-Keyfitz-Pullum equations
- ▶ Matrix-based models
- ▶ Example

What is kinship demography?

- ▶ **Kinship:** “Social relationships that bind individuals together through culturally shared definitions of relatedness on biological, legal, or normative grounds, ultimately constituting family systems’ ’
- ▶ **Family:** “More narrow group of kin given special privilege which, among other things, organize the provision of support, socialization, and social placement of its members.’ ’

from Alburez-Gutierrez et al (2022)

https://osf.io/preprints/socarxiv/fk7x9_v1

What is kinship demography?

- ▶ **Kinship demography:** “The study of family networks, their structures and dynamics from a demographic perspective and using demographic methods.’ ’

What does a demographic perspective mean?

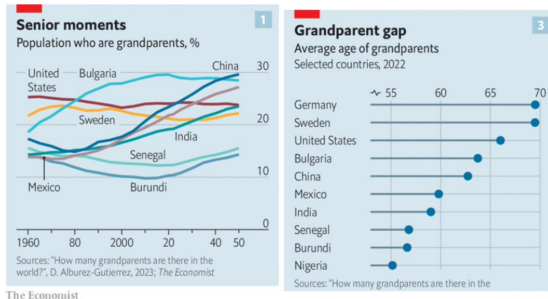
What is kinship demography?

- ▶ Patterns in fertility and mortality imply different kinship structures
- ▶ We saw in Week 5 that, if we ignore migration, a closed population with unchanging mortality and fertility rates has an implicit age structure
- ▶ We can extend this idea to see that a closed population also has implied kinship structures (surviving number of kin of a certain type)

Why is it important?

The age of the grandparent has arrived

The ratio of grandparents to children is higher than ever before. That has big consequences



<https://www.economist.com/international/2023/01/12/the-age-of-the-grandparent-has-arrived>

Why is it important

Estimates from 31 countries show the significant impact of COVID-19 excess mortality on the incidence of family bereavement

<https://www.pnas.org/doi/10.1073/pnas.2202686119>

Why is it important

I'm part of the triple-decker sandwich generation. You probably are as well

Alissa Quart

Mon 1 Dec 2025 16:28 GMT



Share

You've heard of caregivers in the sandwich generation. In this anti-compassion Trump 2.0 era, it's more like a triple-decker sandwich. I experienced this first-hand

<https://www.theguardian.com/us-news/ng-interactive/2025/dec/01/sandwich-generation-trump-parent-caregivers>

Contrast to family demography

“Family demography is a subfield of demography (the scientific study of populations) focused on family and household formation, composition, change, and function. Families are typically defined as those related through blood, marriage, or adoption. Households may be comprised of family members, unmarried cohabiting adults, and/or unrelated persons. ’ ’

From Spence, N.J. (2019). Family Demography. In: Gu, D., Dupre, M. (eds) Encyclopedia of Gerontology and Population Aging. Springer, Cham.
https://doi.org/10.1007/978-3-319-69892-2_319-1

Kinship models

- ▶ Demographic models focus on estimating kinship structures based on a simplified representation of the interaction between fertility and mortality rates
- ▶ Two main types: mathematical or simulation-based
- ▶ Today: focus on mathematical and how to implement in R

Why do we need models?

- ▶ Quantifying kinship structures from data is not necessarily straight forward
- ▶ Family ties are not recorded in vital statistics, and linked data are usually unavailable or highly restructure
- ▶ Some aspects of kinship is captured in surveys, but sample sizes are often small, some groups are underrepresented

Goodman-Keyfitz-Pullum Equations

An absolute banger



Theoretical Population Biology

Volume 5, Issue 1, February 1974, Pages 1-27

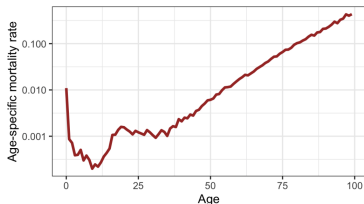


Family formation and the frequency of various kinship relationships

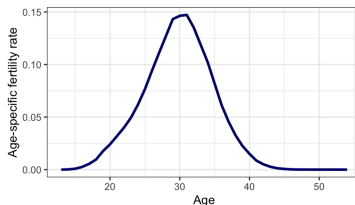
Leo A. Goodman, Nathan Keyfitz, Thomas W. Pullum

Intuition

If you know the mortality rates by age and the fertility rates by age in a population, then you can estimate expected number of different types of kin (e.g. daughters, mothers, etc) still alive



X



Note that GKP focus on female populations only

“A fixed set of age-specific rates implies the probability that a girl aged a has a living mother and great-grandmother, as well as her expected number of daughters, sisters, aunts, nieces, and cousins”

Daughters

$B_1(a)$ is the expected number of living daughters to a woman aged a

$$B_1(a) = \int_{\alpha}^a m(x)l(a-x)dx$$

- ▶ $m(x)$ are the (female) fertility rates of mothers
- ▶ $l(a-x)$ are the survival probabilities of daughters
- ▶ α is start of reproductive lifetime

Daughters

In discrete form:

$$B_1(a) \approx \sum_{\alpha}^a m(x) l(a - x)$$

Example

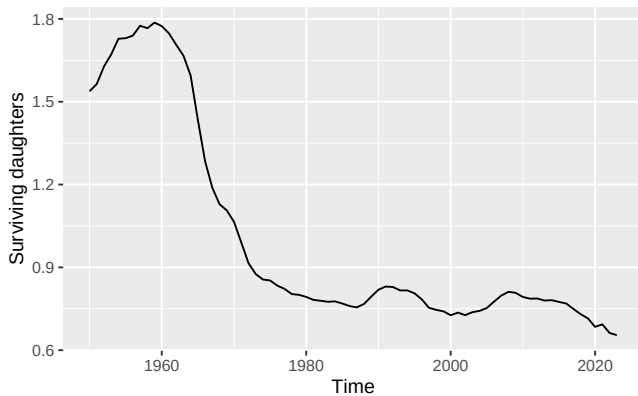
Consider a 60 year old woman in 2023 in Canada. What's the expected number of surviving daughters?

$$B_1(60) = m(10) \cdot l(50) + m(15) \cdot l(45) + m(20) \cdot l(40) + \dots$$

Using 2023 rates this is equal to 0.655.

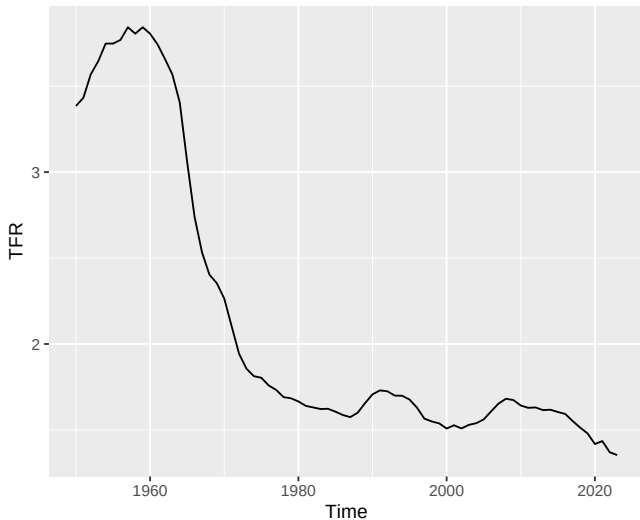
Example

Over time:



Note the synthetic cohort assumption!

Compare to fertility rates



What does surviving daughters tell us that TFR doesn't?

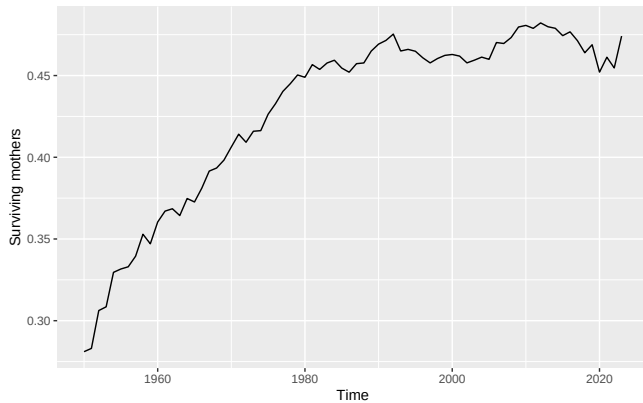
Mothers

$M_1(a)$ is the probability of having a living mother:

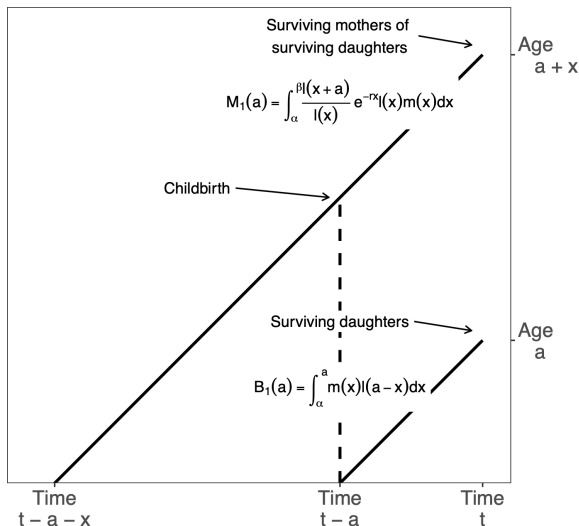
$$M_1(a) = \int_{\alpha}^{\beta} \underbrace{\frac{l(x+a)}{l(x)}}_{\text{prob of surviving from } x \text{ to } a+x} \times \underbrace{W(x)}_{\text{age distribution of mothers}} dx.$$

- ▶ $W(x)$ is the age distribution of mothers. If we assume a stable population, this is $e^{-rx}l(x)m(x)$
- ▶ r is the stable growth rate
- ▶ $m(x)$ and $l(x)$ are (female) fertility rates and survival probabilities as before

Canadian mothers



Lexis diagram



(from Diego's workshop)

Turtles all the way down

e.g. Grand-daughters

Number of surviving grand-daughters to woman aged a :

$$B_2(a) = \int_{\alpha}^a m(x) \int_{\alpha}^{a-x} l(y) m(y) l(a-x-y) dy dx$$

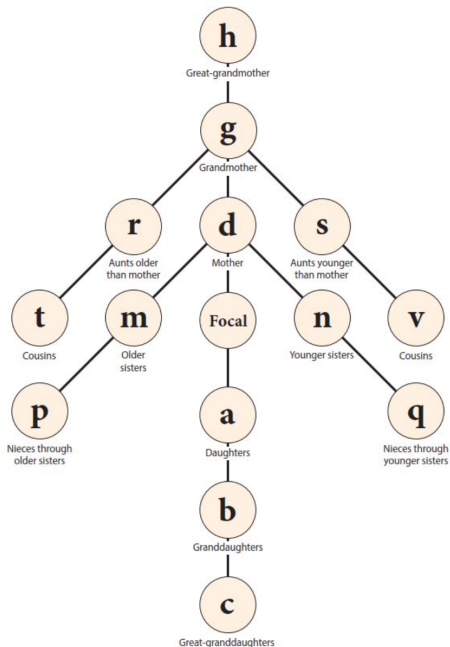
- ▶ Similar logic to get grandmothers
- ▶ Can also get sisters, aunts, cousins

Matrix-based formulations

Enter the matrix

- ▶ Nested integrals get complex fast
- ▶ And what about time varying rates?
- ▶ What about two sex models?
- ▶ Caswell (2019+): takes the Goodman, Keyfitz, Pullum equations, and turns them into a matrix projection model
- ▶ Extends the idea of the cohort component projection model
- ▶ E.g. Treat the children and parents as two different, but connected, populations

Kinship structure



General form

$$\underbrace{\mathbf{k}(x+1)}_{\substack{\text{age structure of kin} \\ \text{at Focal's age } x+1}} = \underbrace{\mathbf{U}\mathbf{k}(x)}_{\substack{\text{ageing and survival} \\ \text{of existing kin}}} + \underbrace{\begin{Bmatrix} \mathbf{0} \\ \mathbf{F}\mathbf{k}^*(x) \end{Bmatrix}}_{\substack{\text{new kin members} \\ \text{added to the population}}}.$$

- ▶ $\mathbf{U}$ is a matrix with survival probabilities in the subdiagonal
- ▶ $\mathbf{F}$ is a matrix with fertility rates in the first row

Example: daughters

$$\underbrace{a(x+1)}_{\text{age structure of daughters at Focal's age } x+1} = \underbrace{Ua(x)}_{\substack{\text{ageing and survival} \\ \text{of existing daughters} \\ a(0) = \mathbf{0}}} + \underbrace{Fe_x}_{\text{new daughters (subsidy)}}$$

- ▶ The second line refers to the the distribution of daughters at the Focal's birth (which is zero)

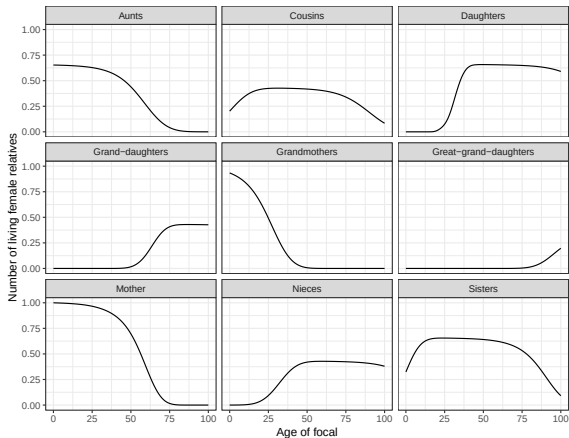
Example: mothers

$$\underbrace{d(x+1)}_{\text{age structure of mothers at Focal's age } x+1} = \underbrace{Ud(x)}_{\text{ageing and survival of existing mothers}} + \underbrace{0.}_{\text{new mothers (subsidy)}}$$

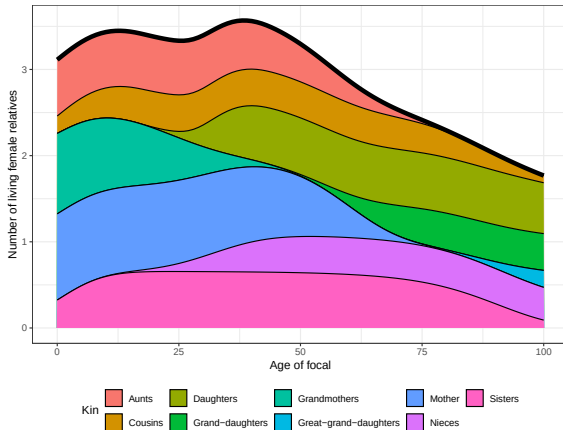
$d(0) = \pi.$

- Second line is the distribution of mothers at Focal's birth - π is the initial distribution (distribution of ages of mothers in the populations)

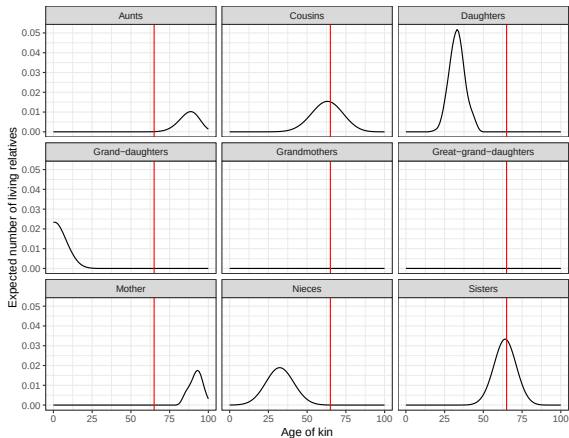
Example: Canada, using 2023 rates



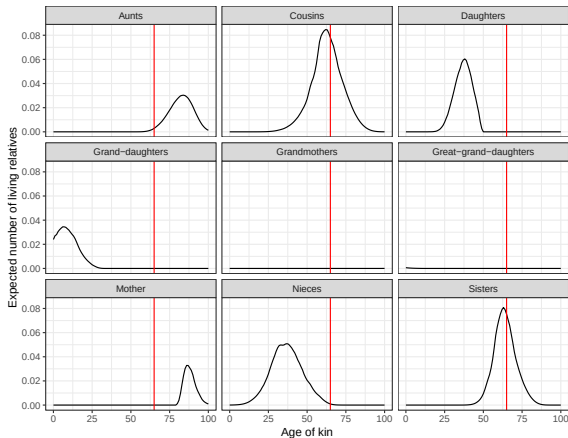
Total kin



Age of kin when ego is 65



Age of kin when ego is 65 using time varying rates



Kinship models in the wild

Original Investigation

FREE

Youth Experiencing Parental Death Due to Drug Poisoning and Firearm Violence in the US, 1999-2020

Benjamin-Samuel Schlüter, PhD¹; Diego Alburez-Gutierrez, PhD²; Kirsten Bibbins-Domingo, PhD, MD, MAS^{3,4};

Monica J. Alexander, PhD^{1,5}; Mathew V. Kiang, ScD⁶

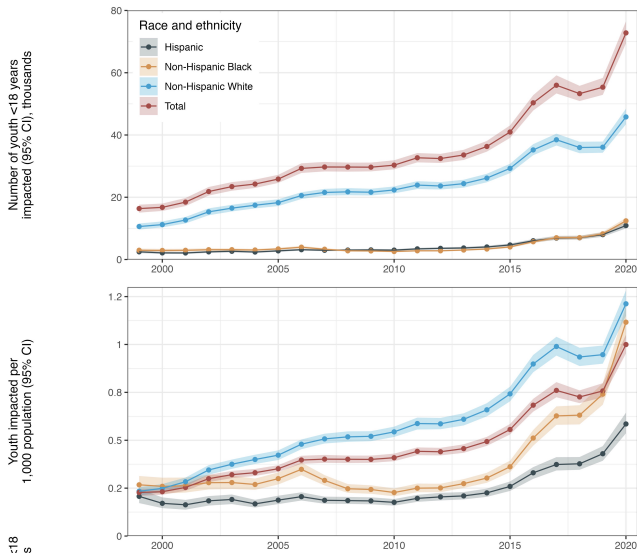
Overview

- ▶ The United States is experiencing dual and overlapping public health crises: drug poisonings and firearm deaths
- ▶ Since 1999, over 1 million people have died due to drug overdoses, and more than 750,000 by firearms
- ▶ The majority of these deaths occur to those aged 15-54, a period when many have young children
- ▶ Drug and firearms deaths have also increased for youths (<18 years); drug poisonings now the third leading cause of death
- ▶ Dual burden for youth: increased mortality, and losing a parent

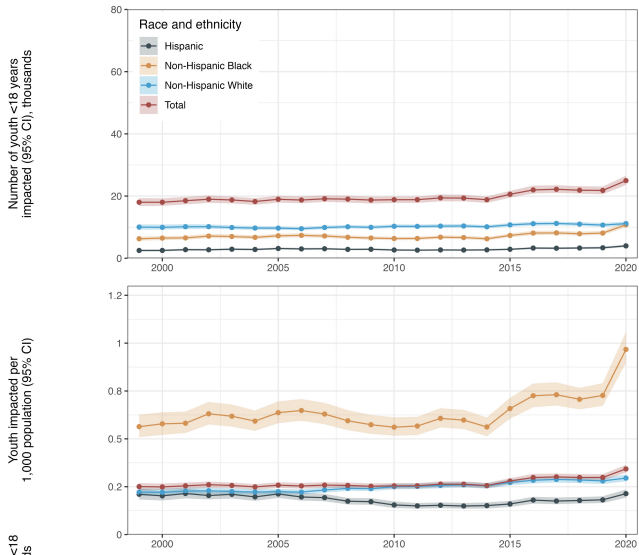
What did we do

- ▶ Extended Matrix-based populations to allow us to track deaths of parents by cause (drugs, firearms, all other causes)
- ▶ Ran separately for three largest race/ethnic groups, and total populations

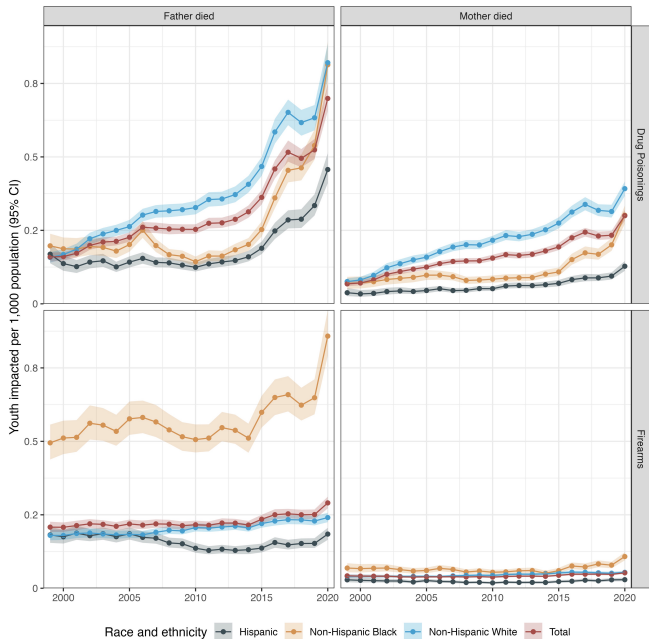
Results - Overdoses



Results - Firearms



Results - by parent



Takeaways

- ▶ An estimated more than 1 million youth lost a parent due to drugs or firearms over the period 1999-2020
- ▶ In 2020, drugs and firearms accounted for almost a quarter of all parental deaths (double the 1999 level)
- ▶ Black youth experienced a disproportionate burden of parental deaths, primarily due to firearm deaths among fathers
- ▶ A whole generation of youth who have lost a parent to traumatic causes
- ▶ Large racial differences of early parental loss, cumulative disadvantage and weathering over the life course
- ▶ Shifting policy to think about the indirect impact on families (US Surgeon General 2024)

Limitations

- ▶ These are model-based estimates!
- ▶ Definition of a parent, assumption of two parents
- ▶ Homogeneity of fertility
- ▶ Race classification, no multi-racial groups

We did validation and sensitivity tests, but potential for future methodological work